

Target BCA

VUG (5)-BCA (C-5001)

2021-24 (VOC)

Full Marks : 70

Time : 3 hours

Answer from **both** the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions: 10×4

1. Define internet and write down about advantages of the internet.
2. What is internet communication ? What are the different types of internet communication ?
3. What is javascript ? What are the advantages of the javascript ? Explain.

(Turn Over)

(2)

4. What is operator in javascript ? What are the types of operator in javascript ? Explain.
5. Write about the XML and what are the building blocks of XML document. Also write a XML program for the 2 students.
6. (a) What is the difference between XML and HTML ?
(b) What is the difference between XML and SGML ?
7. What is jsp ? Discuss about jsp tags with examples.
8. Write about PHP. What are its applications and features ? Write some string functions of PHP.

GROUP—B

Answer *all* questions:

3 × 10

9. What is internet domain ?
10. Write about the TCP.
11. What is array in javascript ?
12. Write about page directive in jsp.
13. Write about the static web page.
14. Write some uses of internet.
15. What do you mean by download and upload operations in internet ?
16. What are the functions of E-mail ?
17. Write about client and server.
18. Write about PHP variable.

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SECTION—A

Answer any *four* questions from the following :

10 × 4

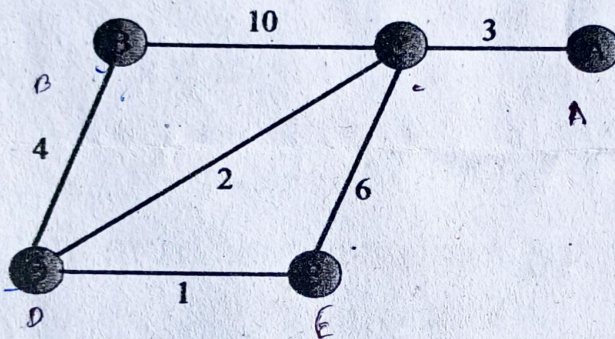
1. (a) What are various basic Asymptotic classes ? Explain Big- O, Big- Ω and Big- Θ notation with examples. 5
- (b) Define Space Complexity and Time Complexity of an algorithm and compute the Time Complexity of Fibonacci Numbers Algorithm. 5

(Turn Over)

2. Sort the following keyword 'ALGORITHM' by applying Quick Sort method.

3. Explain Divide and Conquer technique with the help of any general algorithm.

4. Apply Prim's and Kruskal's algorithm to find Minimum Cost Spanning Tree for the graph given below :

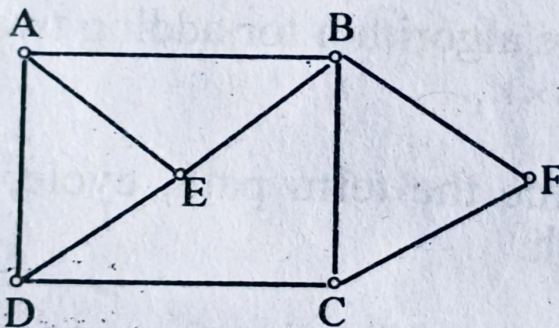


5. Examine the following recurrence relation :

(i) $T(n) = T(n-1) + n^4$

(ii) $T(n) = T(n/4) + T(n/2) + n^2$

6. Explain 'greedy algorithm'. Write its pseudo code to prove a general greedy-choice property of an algorithm.
7. Differentiate between Binary Tree and BST with the help of an example. Construct the Binary search tree for the following elements:
32, 48, 15, 31, 49, 22, 52, 29, 79, 81, 75, 23, 14, 29, 2
8. Write Depth First Search (DFS) Algorithm. Also traverse the following graph using DFS from node A.



SECTION-B

Answer *all* questions from the following :

3 × 10

9. What are Best, Average and Worst Cases ?
Explain.

(4)

$$32 = 2 \times 2 \times 2 \times 2 \times 2$$

$$16 = 2 \times 2 \times 2 \times 2$$

10. Define Big- Θ notation by using basic definition, show that $7n^2 + 8n - 9 = \Theta n^2$.
11. Write Euclid's Algorithm to find GCD of two positive numbers.
12. Apply above algorithm to GCD (325, 95). Show all intermediate steps. Perform Complexity Analysis of above algorithm.
13. Sort the following list using Bubble Sort and show intermediate steps : 30, 8, 7, 14, 20, 28, 10, 6.
14. Write algorithm for adding two matrices of $m \times n$.
15. Define the term path, cycle, complete graph.
16. Solve the following recurrence relation using recurrence tree method :

$$T(n) = 4T(n/2) + n$$

$$5 \times 5 \times 5 \times$$

$$\begin{array}{r} 125 \\ 5 \\ \hline \end{array}$$

$$675$$

$$13251$$

$$\begin{array}{r} 5 \overline{) 325} \\ 5 \overline{) 1625} \\ 5 \overline{) 8125} \\ \hline \end{array}$$

17. List any two applications of BFS/DFS.

18. Arrange following in increasing order :

$$\log_2 n, n \log_2 n, n^2, 5n+7$$

VUG (5)-BCA (C-5003)

2021-24 (VOC)

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed. *analyse*

The questions are of equal value.

Candidates are required to give their answers in their own words as far as practicable.

SECTION-A

Answer any *four* questions:

1. What is Linux ? What is the role of the free software foundation in the development of Linux ? Who developed Linux Kernel ?
2. Explain the different features of Linux operating system.
3. Explain the Kernel architecture of Linux.
4. Explain what are the main differences between Linux & UNIX.

(Turn Over)

5. Explain the process control of Linux System.
6. Explain the different steps of Process Management in Linux operating system.
7. Explain five commands in Linux operating system.

SECTION-B

8. Answer *all* questions of the following:
 - (a) What are the various Linux distributions ?
 - (b) What is Device Driver ?
 - (c) File system hierarchy of Linux.
 - (d) Use `chmod-w` and then try to create and remove a file in current directory.
 - (e) What is file & record locking ?
 - (f) What is Shell ?

- (g) What is thread ?
- (h) Define Ed Editor.
- (i) Explain Sed.
- (j) Explain 3 Awk built in variable names.

VUG(5)-BCA (C-5004)

2021-24 (VOC)

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GROUP—A

Answer any *four* questions

10 × 4

1. Find a real root of equation

$f(x) = x^3 + x^2 - 1 = 0$ using iteration method.

2. The equation $x^6 - x^4 - x^3 - 1 = 0$ has one real root between 1.4 and 1.5. Find the root to four decimals by false position method.

3. Using Simpsons 3/8th rule

$3x^2 + 2x$

1.5 0.1250

(Turn Over)

$f(1.5) = 3(1.5)^2 + 2(1.5) = 6.75 + 3 = 9.75$

(2)

Find $\int_0^6 \frac{dx}{(1+x)^2}$.

4.

$x :$	5	6	9	11
$y :$	12	13	14	16

Find the value of y when $x = 10$ by Lagrange's formula.

5. Estimate the number of students who obtained marks between 40 and 45.

Marks	No. of students
30-40	31
40-50	42
50-60	51
60-70	35
70-80	31

6. Solve

$$2x_1 + 3x_2 + x_3 = 9$$

$$x_1 + 2x_2 + 3x_3 = 6$$

$$3x_1 + x_2 + 2x_3 = 8$$

(3)

by Gauss elimination method.

7. Apply R-K method to solve

$\frac{dy}{dx} = x + y, y(0) = 1$ and find the value of y at $x = 0.2$ taking $n = 0.1$ by order 4.

8. Solve

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

$$x = \frac{85 - 6y + z}{27}$$

$$y = \frac{72 - 6x - 2z}{15}$$

by Gauss Seidal method. $z = \frac{110 - x - y}{54}$

GROUP—B

3 × 10

Answer *all* questions.

9. What is error ?

10. What is Absolute, Relative and percentage error ?

11. What is interpolation ?

12. What is Taylor's series ?

$$2 \times \frac{9}{5} - 6$$

$$\frac{14}{5} - 6$$

$$\frac{14-25}{5} = -\frac{11}{5}$$

$$\frac{11}{5} \div \frac{14-16}{5}$$

(4)

13. Prove that $\Delta = E - 1$.

14. What is Trapezoidal rule ?

15. What is Euler's method ?

16. What is Bisection method ?

17. What is R-K second order ?

18. What is divided difference ?

VUG(5)-BCA (C-5007)

2021-24 (VOC)

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GROUP—A

Answer any *four* questions. 10×4

- ✓ 1. Write about the business to consumer (B2C) business models.
- ✓ 2. How the internet and the web change business ?
- ✓ 3. Write and explain the e-commerce site tools.

(Turn Over)

4. Write and explain the hardware for an e-commerce site.
5. Write the policy and laws for e-commerce security.
6. (a) Write about B2C arena.
(b) B2B payment systems.
7. Write about ethical, social and political issues in e-commerce.
8. (a) Write about privacy and information rights.
(b) Write about Intellectual property rights.

GROUP—B

Answer *all* questions.

3 × 10

9. What is e-commerce ?
- ✓10. What is B2B ?

- ✓ 11. What is WWW ?
- ✓ 12. What is Encryption ?
- ✓ 13. Write about e-commerce Payment Systems.
- ✓ 14. Write the advantages of e-commerce.
- ✓ 15. Write the disadvantages of e-commerce.
- ✓ 16. What is firewalls ?
- ✓ 17. What is internet ?
- ✓ 18. What is ethical issues ?

Target BCA